

COLDPOD · DESIGN DECISION RECORD

ColdPod TRL 2 review decisions

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	CPD-DDR-001	Version	0.2
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/coldpod		
Source	docs/decisions/0001-trl2-review-decisions.md @ b4afd8f		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); item O2 now decided, see CPD-DDR-002
0.1	2026-09-25	Amish Chadha	Record Amish's decisions on the TRL 2 review items and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted for items D1 to D11; item O2 accepted through CPD-DDR-002; item O1 remains proposed

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-25, /populate) listed eleven items as "Proposed, awaiting Amish". On 2026-09-25 Amish reviewed the review points for every portfolio repo and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." Every item that carried a recommendation is therefore decided in favor of that recommendation. The one item without a recommendation stays open. TRL 4 work is on hold by the same instruction.

■ Options considered

The options for each item are those listed in docs/REVIEW.md (TRL 2 session) and in CPD-PRC-001 v0.2, Key design choices.

■ Decision

Table 1. Decided items.

#	Item	Decision
D1	Phase-change material	5 °C organic PCM (RT 5 HC class), rather than water ice or a 3 °C PCM, to prevent freezing. Decided by Amish, 2026-09-25: go with recommendation.
D2	Cold path	Gravity thermosiphon as a one-way thermal link, with a mechanical disconnect as the fallback, rather than bolting the Peltier module to the liner. Decided by Amish, 2026-09-25: go with recommendation.
D3	Evaporator position	On the outer face of the PCM, not on the liner. Decided by Amish, 2026-09-25: go with recommendation.
D4	Battery	LiFePO4 4S1P, 12.8 V 6 Ah (76.8 Wh), rather than 4S2P (153.6 Wh). Decided by Amish, 2026-09-25: go with recommendation.
D5	R6 shortfall	Accept about 16 h off-grid at 43 °C and rely on vehicle or solar power for long hot trips, rather than a larger battery or thicker insulation. Decided by Amish, 2026-09-25: go with recommendation.
D6	R12 shortfall (mass)	Relax the empty-mass target to 5.5 kg, rather than cutting shell and liner mass. Decided by Amish, 2026-09-25: go with recommendation.
D7	Insulation	25 mm vacuum-insulated panels, with a polyurethane foam version documented as a low-cost variant. Decided by Amish, 2026-09-25: go with recommendation.
D8	Connectivity	Local alarms plus Bluetooth Low Energy only in the first build; cellular or LoRa later. Decided by Amish, 2026-09-25: go with recommendation.
D9	Alarm defaults	Warn after 10 min outside 2 to 8 °C; alarm at once at 0 °C or lower on the payload probe; early freeze warning at 1 °C on the liner probe; adjustable per product. Decided by Amish, 2026-09-25: go with recommendation.
D10	Budget	Raise <code>budget_usd</code> from \$250 to \$300. Decided by Amish, 2026-09-25: go with recommendation.
D11	First user	Outreach vaccinators first (sets the 43 °C design case), then travellers with insulin. Decided by Amish, 2026-09-25: go with recommendation.

Notes on the decided items:

- **D5.** Requirement R6 in CPD-REQ-001 v0.3 is redefined from 24 h to 16 h or more at 43 °C off-grid, with longer hot trips run from external power under R7.
- **D6.** Requirement R12 in CPD-REQ-001 v0.3 is relaxed from 5.0 kg to 5.5 kg empty. The TRL 3 mass estimate is 5.60 kg (CPD-CAL-001, K2), so R12 is still not met; see `docs/REVIEW.md`.
- **D10.** `budget_usd` in `project.yaml` is now 300. R15 is redefined to \$300. The TRL 3 BOM totals \$295 (CPD-CAL-001, L2).
- **D2 and D3.** At TRL 3 the evaporator on the outer face of the PCM is sized as an aluminium can around the PCM jacket, fed by a two-phase loop thermosiphon (CPD-CAL-001, sections G and H). This sizes the decided choice; it is listed for confirmation in `docs/REVIEW.md`.
- **SwapCell.** The TRL 2 review did not propose the SwapCell 48 V pack (about 468 Wh and 2.8 kg, far larger than ColdPod needs). The portfolio decisions of 2026-09-25 on the SwapCell interface (v0.3 items: wake without CAN, charge-while-discharging mode, latch vibration rating) and on pricing shared packs once therefore do not change the ColdPod design or BOM.
- **Pitch and problem lines.** The review recommended no change to the `pitch` or `problem` wording, so `project.yaml` keeps them.

Table 2. Items left open by this record, and their state after CPD-DDR-002.

#	Item	State
O1	Co-design partner (an immunization program, a diabetes association or a humanitarian logistics group)	Proposed, awaiting Amish. No recommendation was made; the portfolio decision is to pick partners per area later
O2	New TRL 3 proposals: lid pack cold path, liner freeze cut-out, mass, evaporator can and loop thermosiphon, Peltier class and drive, vehicle input headroom, budget margin	Decided by Amish, 2026-09-25: go with recommendation. Recorded item by item, with what changed in the repo, in CPD-DDR-002 (<code>0002-recommendations-accepted.md</code>). The thermosiphon tilt limit carried no separate recommendation; it stays a test question for after TRL 3

■ Consequences

- CPD-PRB-001, CPD-PRC-001 and CPD-REQ-001 are revised to v0.3 to show the decisions: design choices 1 to 9 of the precis are no longer "proposed", R6, R12 and R15 carry the new targets, and the first user is named.
- `project.yaml` carries `budget_usd: 300`.
- The partner question (O1) stays open in CPD-PRB-001.